Spatio-Temporal Analytics
Submission Report

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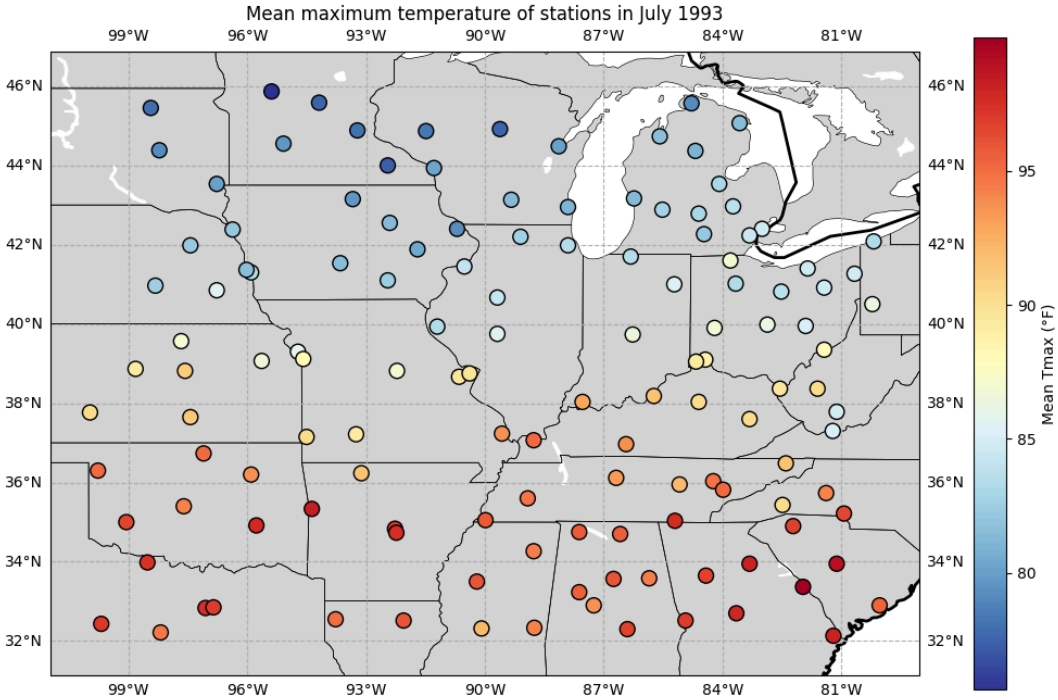
1. Introduction

The investigated NOAA dataset contains daily weather observations from 137 weather stations across the United States between 1990 and 1993. It has 729410 rows and 10 columns. Each row represents a record of one of four measurements at a single station on a single day with the following variables:

- **id**: the station ID
- **lon, lat**: the geographic coordinates of the station in degrees
- **day, month, year, date, julian**: temporal identifiers
- **proc**: the measurement type (Tmax, Tmin, TDP, Precip)
- **z**: the recorded value for the specified measurement

The data was preprocessed to filter the measurement type to the maximum temperature measurements (z = Tmax in °F) for July 1993. A single Tmax value was missing and was removed, producing a dataframe with 4122 observations from 133 stations.

The data was visualised using the cartopy library to show the locations of the stations and the mean maximum temperature for each of them in July 1993. This shows that station coverage was not uniform across the area. It can also be seen that maximum recorded temperatures appear to follow a trend: on average, colder maximum temperatures were measured in the northern United States, with average measurements steadily increasing going south.



Southern stations recorded higher average maximum temperatures. The data was restructured into a labelled dataset:

$$D = \{(\text{lon}_i, \text{lat}_i, t_i, Z_i)\}_{i=1,\ldots,m}$$

Where Z_i is the observed Tmax at location $s_i = (lon_i, lat_i)$ on day t_i and m is the number of known observations.

2. Ordinary Least Squares Regression Model

Model Definition:

$$Z(s_i, t_i) = \beta_0 + \beta_1 \text{lat}_i + \beta_2 \text{lon}_i + \beta_3 t_i + \varepsilon(s_i, t_i), \quad i = 1, \ldots, m$$

where:

- $Z(s_i, t_i)$ is the observed Tmax at station $s_i = (lon_i, lat_i)$ on day t_i;
- β_0 is the intercept;
- $\beta_1, \beta_2, \beta_3$ are the regression coefficients for latitude, longitude and day respectively;
- $\varepsilon(s_i, t_i)$ is the error term at location (s_i, t_i).

OLS Estimator:

$$\hat{\beta}_{OLS} = \arg\min_{\beta} \sum_{i=1}^{m} \left(Z(s_i, t_i) - x_i^\top \beta\right)^2 = (X^\top X)^{-1} X^\top Z$$

For OLS to be valid, the following assumptions are required:

- The model is linear;
- The mean of errors is zero;
- Homoscedasticity (constant variance);
- Independence (errors are uncorrelated).

Looking at the estimated coefficients in the OLS model summary, we can see that latitude is the strongest predictor by a large margin. A one degree increase in latitude is associated

with a 1.558 degree drop in temperature, compared to 0.032 and 0.072 changes for longitude and day respectively. Latitude's t-statistic (-83.6) is much larger in magnitude than the others' (2.5 and 8.8 respectively), confirming it dominates the model. This points towards there being:

- A strong north-south temperature gradient (colder maximum temperatures at higher latitudes);
- A very weak east-west gradient (maximum temperature increases very slightly moving eastward);
- A weak gradual warming through July (gentle gradual increase in maximum temperature throughout the month).

OLS Regression Results

Dep. Variable:	z	R-squared:	0.633
Model:	OLS	Adj. R-squared:	0.633
Method:	Least Squares	F-statistic:	2365.
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	0.00
Time:	12:14:59	Log-Likelihood:	-12224.
No. Observations:	4122	AIC:	2.446e+04
Df Residuals:	4118	BIC:	2.448e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	150.5083	1.334	112.850	0.000	147.893	153.123
lat	-1.5581	0.019	-83.583	0.000	-1.595	-1.522
lon	0.0322	0.013	2.518	0.012	0.007	0.057
day	0.0717	0.008	8.764	0.000	0.056	0.088

Omnibus:	67.145	Durbin-Watson:	0.946
Prob(Omnibus):	0.000	Jarque-Bera (JB):	89.618
Skew:	-0.212	Prob(JB):	3.46e-20
Kurtosis:	3.584	Cond. No.	1.80e+03

The model explains 63.3% of the variance in Tmax (R^2= 0.633) with the three covariates, indicating that the dominant structure in the data is the smooth, large scale trend associated with latitude, rather than local or irregular variation. 36.7% of the variance not being captured is too high for this to be random noise however. This suggests that the OLS model is not capturing the complete spatio-temporal structure. In addition, the Durbin-Watson value (0.946 < 2) indicates that there is a potential correlation between errors at nearby stations, which would violate the assumption of independence.

3. Estimating the Covariance Function

A Gaussian covariance function was used to model the spatio-temporal dependence structure:

$$C_{\boldsymbol{\theta}}((s_i,t_i),\,(s_j,t_j)) = \sigma^2 \exp\left(-\frac{(\text{lon}_i-\text{lon}_j)^2}{a^2} - \frac{(\text{lat}_i-\text{lat}_j)^2}{b^2} - \frac{(t_i-t_j)^2}{c^2}\right) + \tau^2\,\mathbf{1}_{\{i=j\}}$$

The parameters of this function are:

- σ^2: the underlying variance explained by spatio-temporal components, representing the maximum covariance between two observations;
- a: the degree to which covariance decays longitudinally;
- b: the degree to which covariance decays latitudinally;
- c: the degree to which covariance decays temporally;
- τ^2: the variance unexplained by spatio-temporal components, representing random noise.

Parameters	**Increase**	**Decrease**
σ^2	Stronger spatio-temporal dependence	Weaker spatio-temporal dependence
a	Correlation persists over larger longitudinal distances	Correlation decays more quickly east to west
b	Correlation persists over larger latitudinal distances	Correlation decays more quickly north to south
c	Correlation persists over larger time periods	Correlation decays more quickly over time
τ^2	More random noise	Less random noise

This covariance model was fitted to the OLS residuals to test whether the remaining variation had spatial and temporal structure. Due to the computational cost of the $O(n^3)$ matrix factorisation required to evaluate the Gaussian log-likelihood, the parameters were estimated on a random subsample (size 1000) of D rather than the full dataset. The Gaussian likelihood is defined as:

$$L(\theta) = |C(\theta)|^{-1/2} \exp\left(-\frac{1}{2}\mathbf{r}^\top C(\theta)^{-1}\mathbf{r}\right)$$

so the log-likelihood is:

$$\ell(\theta) = -\frac{1}{2}\log|C(\theta)| - \frac{1}{2}\mathbf{r}^\top C(\theta)^{-1}\mathbf{r}$$

where $\mathbf{r}$ are the residuals of the OLS model and $C(\boldsymbol{\theta})$ is the covariance matrix. Using 'scipy.optimize.minimize' to maximise the log-likelihood, evaluated at each step using 'multivariate_normal.logpdf', the estimated parameters are:

- σ^2 = 15.73
- a = 5.22 (autocorrelation decays over 5 - 6 degrees of longitude)
- b = 3.00 (autocorrelation decays over roughly 3 degrees of latitude)
- c = 1.18 (autocorrelation decays over 1 - 2 days)
- τ^2 = 4.15

At 3.79, the signal-to-noise ($\frac{\sigma^2}{\tau^2}$) shows that roughly 80% of the residual variance was spatially structured and only 20% was random noise. This confirms that the independence assumption in the OLS model was violated. Standard errors from the OLS are likely underestimated, meaning t-statistics and p-values would be overstated.

4. Prediction Grid

A 20 x 20 spatial grid was constructed by splitting [80°W, 100°W] into 20 evenly spaced longitudes and [32°N, 46°N] into 20 evenly spaced latitudes. Six days spread 6 days apart across July 1993 were selected (1st, 7th, 13th, 19th, 25th, 31st). The prediction grid was defined as:

$$G = \{(\text{lon}, \text{lat}, t) : (\text{lon}, \text{lat}) \in \text{spatial grid},\ t \in \{1, 7, 13, 19, 25, 31\}\}$$

This comprises 400 spatial locations and 6 time steps, giving a total of 2400 space-time locations.

5. Covariance-based Universal Kriging

The universal Kriging predictor gives the Best Linear Unbiased Predictor (BLUP) for Tmax at any unobserved spatio-temporal location. It improves on the OLS trend by incorporating the spatial-temporal dependence structure estimated with the Gaussian covariance function previously. It is defined as:

$$\hat{Z}(s_0, t_0) = x_0^\top \hat{\beta}_{GLS} + c_0^\top C_\varepsilon^{-1}(Z - X\hat{\beta}_{GLS})$$

Where:

- $x_0^\top \hat{\beta}_{GLS}$ is the trend component
- $c_0^\top C_\varepsilon^{-1}(Z - X\hat{\beta}_{GLS})$ is the Kriging correction

The trend coefficients are estimated by GLS rather than OLS so that spatial correlation in the errors can be captured. The GLS estimator is:

$$\hat{\beta}_{GLS} = (X^\top C_\varepsilon^{-1} X)^{-1} X^\top C_\varepsilon^{-1} Z$$

This is more efficient than the OLS estimator because it down-weights observations that are highly correlated with their neighbours and up-weights those carrying more independent information.

Cross-covariance:

The vector c_0 is the cross-covariance between the prediction location (s_0, t_0) and all 4122 observations in D, computed from the fitted Gaussian covariance function without τ^2:

$$c_0^{(j)} = \sigma^2 \exp\left(-\frac{(\text{lon}_0 - \text{lon}_j)^2}{\hat{a}^2} - \frac{(\text{lat}_0 - \text{lat}_j)^2}{\hat{b}^2} - \frac{(t_0 - t_j)^2}{\hat{c}^2} \right)$$

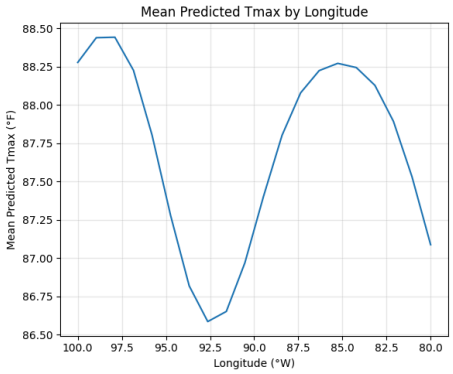
τ^2 is excluded because it represents measurement error at observed stations, so does not apply at new prediction locations.

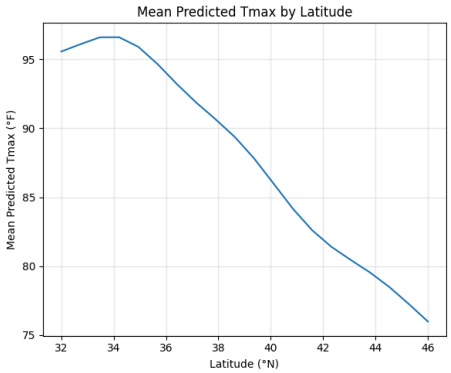
Assumptions:

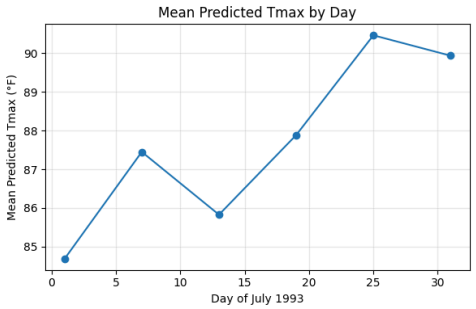
- The residual dependence structure is correctly described by the Gaussian covariance function C_θ with estimated parameters $\hat{\theta}$;
- The residuals follow a multivariate Gaussian distribution with mean zero and covariance C_ε.

Plotting mean predicted Tmax by latitude and longitude shows the following:

- Mean predicted Tmax steadily decreases from 34°N to 46°N by roughly 20°F;
- Mean Tmax variation by longitude is less than 2°F, but does appear to follow a trend, with peaks at 98°W and 85°W, and a trough at 92.5°W.







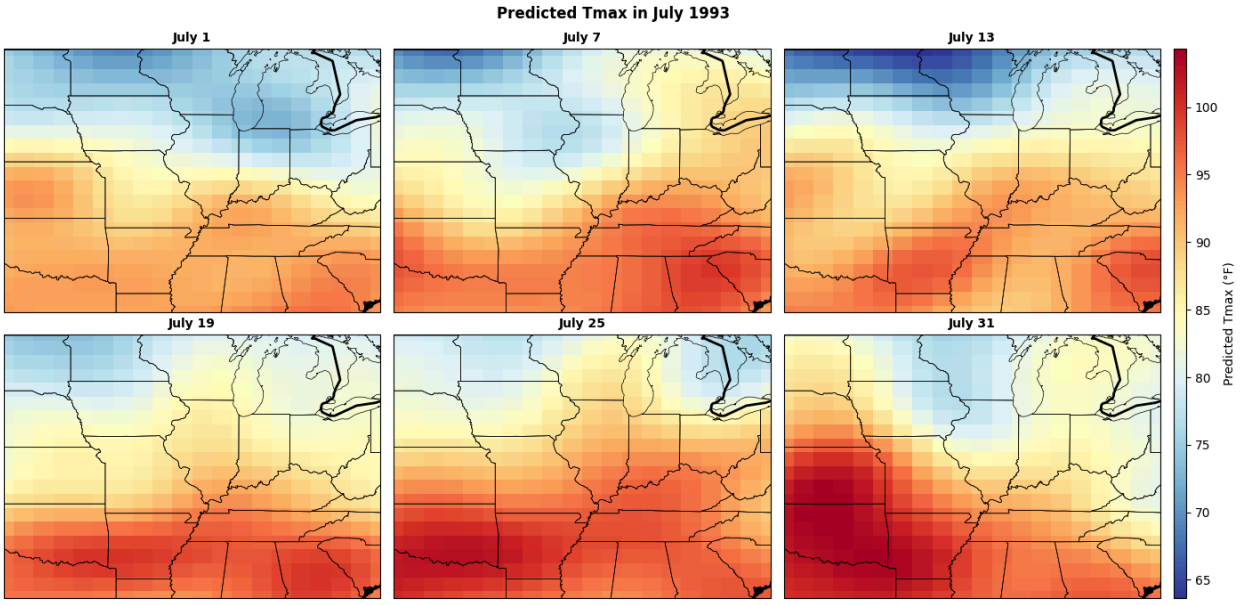
Temporally, mean predicted Tmax increases over July 1993 overall by roughly 5°F, but not consistently.

Kriging Variance:

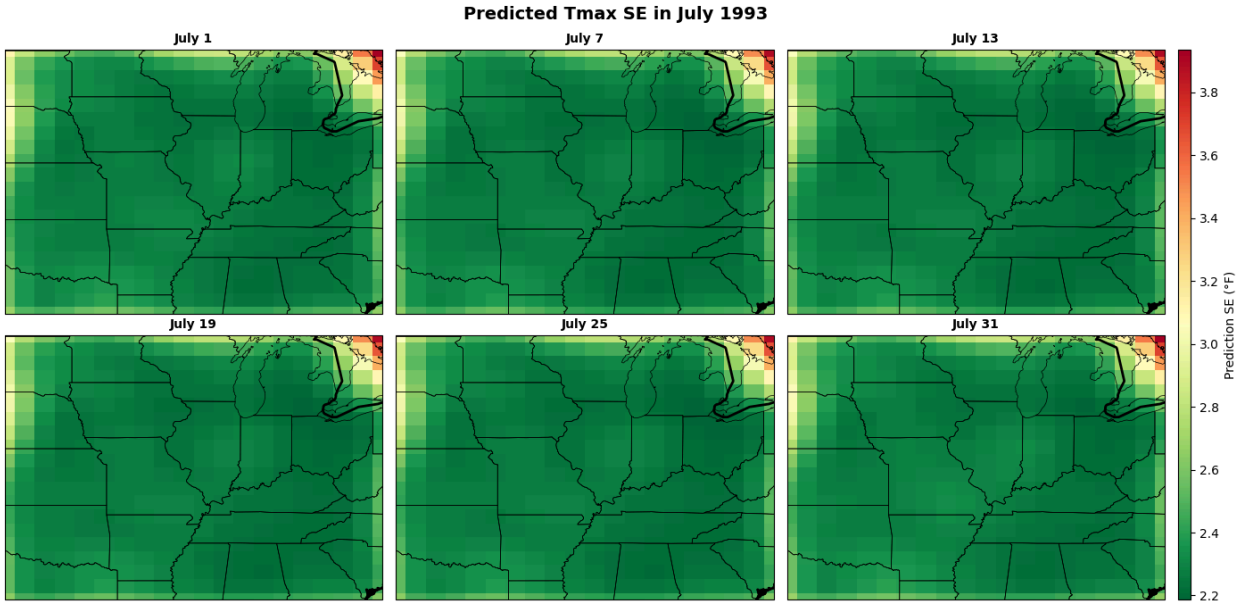
The prediction uncertainty at each grid point is quantified by:

$$\mathrm{Var}(\hat{Z}(s_0,t_0) - Z(s_0,t_0)) = C(0) - c_0^\top C_\varepsilon^{-1} c_0 + u_0^\top (X^\top C_\varepsilon^{-1} X)^{-1} u_0$$

The six-panel prediction map consistently shows a clear north-south temperature gradient throughout the month, expressing local differences well. There is no obvious trend longitudinally or temporally from these.



The prediction standard errors appear to be smallest in the areas where stations are densest (<2.25°F) and increase in pockets where fewer stations are located. This is most evident at the edges of the map and in the north-east section of the plot over the Canadian border (>4.00°F), for which there is no station data in the dataset. That all six of these plots appear to be almost identical suggests that uncertainty is governed by the spatial configuration of the station network rather than time.

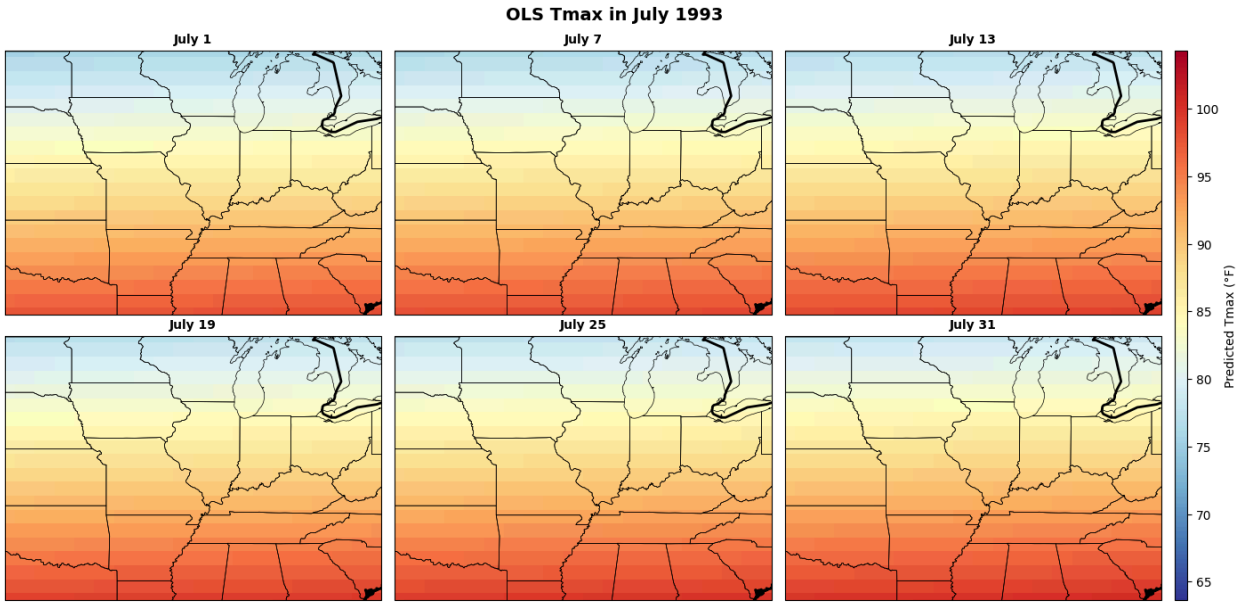


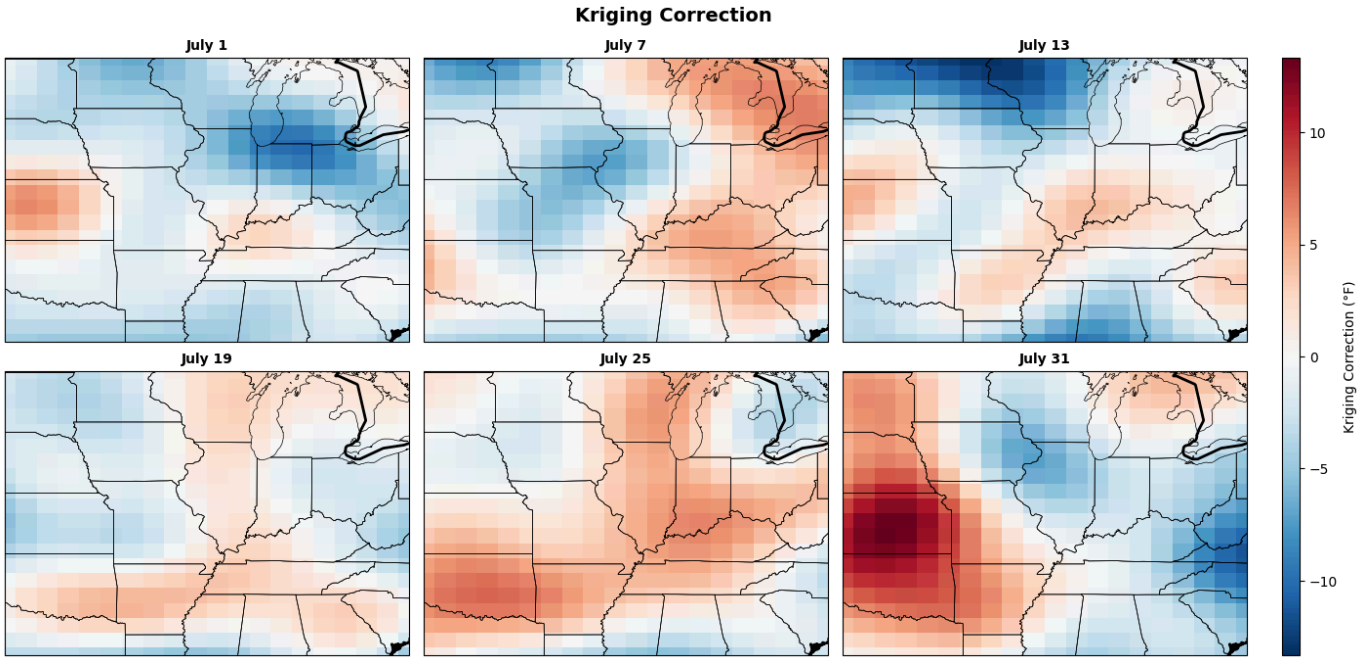
6. Comparison

The OLS trend produces smooth, featureless predictions that vary only as a function of the covariates, with no correction from nearby observations. It is not able to capture local temperature deviations. By pulling predictions toward nearby observed values, Universal Kriging manages to produce a spatially coherent trend with predictions that are better reflected in the real world. OLS gives a single predicted value with standard errors, while Kriging is able to give a prediction and a variance at each location. In this case, since the errors were not independent, OLS was not sufficient. As a BLUP, Kriging was able to optimally fit the covariance structure.

With a signal-to-noise ratio of roughly 80%, it is likely that the covariance model captured genuine spatio-temporal information rather than fitting random noise. From the results, correlation decays over about 5° longitudinally and 3° latitudinally. At around 1.2 days, the temporal correlation is short, reflecting day-to-day weather persistence.

That is not to say that the OLS model is not useful at all. It works well as a good baseline temperature with which to compare. For example, from the correction plot, it can be seen that on 13 July, colder than average temperatures for that area were measured in the north. On 31 July, western stations measured higher temperatures than are usually experienced there.





7. Conclusion:

The Kriging model fits the NOAA data to a much better degree than the OLS model. The OLS model could only explain 63.3% of the variance of Tmax. The estimated covariance function on the residuals of the OLS found parameters which indicated that the unexplained variance comprised substantial spatio-temporal structure. The Kriging model explicitly accounts for this structure and produces predictions that reflect both the large-scale trend and local dependence, making it the more realistic predictor for spatially correlated data such as this.

GenAI Statement:

Google Gemini was used to suggest the cartopy library used in Section 1, the OLS method used in Section 2 and the TwoSlopeNorm method used in the comparison of Section 5. All work is my own and was done without assistance from any other person or generative AI.